DS 3001: HW 5: Linear Models

Question 4:

(1) $\hat{y}_i = b_0 + b_1 z_{i1} + b_2 z_{i2}$

$SSE(b_0, b_1, b_2) = \sum_{i=1}^{n} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})^2$

(2)

partial derivative w/ respect to b_0

$$\frac{\partial SSE}{\partial b_0} = -2\sum_{i=1}^{N} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})$$

w/ respect to b_1:

$$\frac{\partial SSE}{\partial b_1} = -2\sum_{i=1}^{N} z_{i1}(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})$$

w/ respect to b_2:

$$\frac{\partial SSE}{\partial b_2} = -2\sum_{i=1}^{N} z_{i2}(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})$$

$\rightarrow \sum y_i = nb_0 + b_1 \sum z_{i1} + b_2 \sum z_{i2}$

$\sum z_{i1} y_i = b_0 \sum z_{i1} + b_1 \sum z_{i1}^2 + b_2 \sum z_{i1} z_{i2}$

$\sum z_{i2} y_i = b_0 \sum z_{i2} + b_1 \sum z_{i1} z_{i2} + b_2 \sum z_{i2}^2$

(3) let residuals be: $e_i = y_i - \hat{y}_i$

$\sum e_i = 0 \rightarrow \bar{e} = 0$

From b_1 & b_2 equations: $\sum z_{i1} e_i = 0$, $\sum z_{i2} e_i = 0$

so, $e \cdot z_1 = 0$, $e \cdot z_2 = 0$

so, residuals are orthogonal to predictors

(4) Set first derivative to 0:

$\sum_{i=1}^{N} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$

$\sum y_i - Nb_0 - b_1 \sum z_{i1} - b_2 \sum z_{i2} = 0$

Since variables are centered, $\sum z_{i1} = 0$ and $\sum z_{i2} = 0$. The equation simplifies to:

$\sum y_i - Nb_0 = 0 \rightarrow b_0^* = \frac{1}{N} \sum y_i = \bar{y}$

(5) Substitute $b_0 = \bar{y}$ into remaining derivative equations

$\sum z_{i1}(y_i - \bar{y} - b_1 z_{i1} - b_2 z_{i2}) = 0$

$\sum z_{i2}(y_i - \bar{y} - b_1 z_{i1} - b_2 z_{i2}) = 0$

Rearranging

$$\underbrace{\begin{bmatrix} \sum z_{i1}^2 & \sum z_{i1} z_{i2} \\ \sum z_{i1} z_{i2} & \sum z_{i2}^2 \end{bmatrix}}_{\text{by coefficients}} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \underbrace{\begin{bmatrix} \sum z_{i1}(y_i - \bar{y}) \\ \sum z_{i2}(y_i - \bar{y}) \end{bmatrix}}_{\text{by coefficients}}$$

Scanned with CamScanner

(6) Dividing every term by N transforms the sums into sample variances & covariances

Matrix A:

$$A = \begin{bmatrix} var(x_1) & cov(x_1, x_2) \\ cov(x_1, x_2) & var(x_2) \end{bmatrix}$$

- $\frac{1}{N} \sum z_{i1}^2 = var(x_i)$
- $\frac{1}{N} \sum z_{i1} z_{i2} = cov(x_1, x_2)$
- $\frac{1}{N} \sum z_{i1} \bar{y}_1 = cov(x_1, y)$

Vector C:

$$C = \begin{bmatrix} cov(x_1, y) \\ cov(x_2, y) \end{bmatrix}$$

This discovery shows that the optimal coefficients depend entirely on how the variables vary and co-vary. Specifically, b_1 & b_2 are chosen such that they account for the shared variance between predictors. If x_1 & x_2 are perfectly uncorrelated (cov = 0), the matrix A becomes diagonal, and the multiple regression coefficients would simply be the same as individual single linear regression coefficients.